

Preamble

A central ingredient in field theory is diagrammatics. In classical φ^4 theory, there are, to one loop, only two diagrams that determine the critical exponents:



I will briefly review how they determine the exponents below. I can see that they are essentially a matter of counting contributions. However:

- it's not about their *asymptotic* combinatorics, as far as I can tell;
- std. DP problems are often *not* of this form. There, it's usually about summing over *all* diagrams, or many, in a systematic way.

1 Standard φ^4 is not about asymptotic numbers of diagrams

(a) Tadpole

$$\begin{aligned}
 \text{tadpole} &\hat{=} \int \frac{d^d k}{(k^2 + r)} = K_d \int_0^\infty dk \frac{k^{d-1}}{k^2 + r} \\
 &= K_d \frac{1}{2} \int du \frac{u^{d/2-1}}{u + r} = K_d \frac{1}{2} r^{d/2-1} \frac{\Gamma(1 - \frac{d}{2}) \Gamma(1)}{\Gamma(\frac{d}{2})} \\
 &= A r r^{-\varepsilon/2} \boxed{\frac{\Gamma(\varepsilon/2)}{1-d/2}}
 \end{aligned}$$

with $A = \frac{1}{2} K_d$, $\Gamma(1) = 1 = \Gamma(2)$, $\Gamma(1 - \frac{d}{2})(1 - \frac{d}{2}) = \Gamma(2 - \frac{d}{2})$, $\varepsilon = 4 - d$.

Combinatorial factor: $4 \cdot 3 = \underbrace{\frac{4 \cdot 3}{2}}_{\text{loop}} \cdot \underbrace{2}_{\text{external}}$

(b) Bubble (sunset / two-line)

$$\begin{aligned}
 \text{bubble} &\hat{=} \int \frac{d^d k}{(k^2 + r)^2} = K_d \frac{1}{2} \int du \frac{u^{d/2-1}}{(u + r)^2} \\
 &= K_d \frac{1}{2} r^{d/2-2} \frac{\Gamma(2 - \frac{d}{2})}{\Gamma(2) \Gamma(\frac{d}{2})} = A r^{-\varepsilon/2} \boxed{\Gamma(\varepsilon/2)}
 \end{aligned}$$

Combinatorial factor: $\binom{4}{2} \binom{4}{2} \cdot 2 \cdot 4! \cdot \frac{1}{2}$ (legs to form loop · external · spinlet relabel)

So

$$g = 2A r^{-\varepsilon/2} u_4 \quad (\text{sd. nonlinearity}), \quad \Gamma(\varepsilon/2) = \frac{2}{\varepsilon} + \text{h.o.t.}$$

Resulting flow and exponent

$$(\beta) \quad \beta_u = -\varepsilon \cdot 4! g + 36 \cdot 4! g^2 \xrightarrow{\beta_u=0} g^* = \frac{\varepsilon}{36}$$

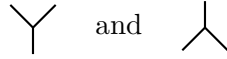
$$(\gamma) \quad Z_r = 1 + 12g \Rightarrow \gamma_r = 12g^* = \frac{1}{3}\varepsilon$$

$$(\nu) \quad \frac{1}{\nu} = 2 - \gamma_r \Rightarrow \nu = \frac{1}{2} \frac{1}{1 - \gamma_r/2} = \frac{1}{2} + \frac{\gamma_r}{4} + \dots = \frac{1}{2} + \frac{1}{12}\varepsilon$$

The exponent is determined by the **ratio** of the integrals and the combinatorial factors as highlighted. This is **not** about asymptotics, just a matter of counting.

2 My current favourite DP problem

Consider the two vertices



and label them as H -vertices with leg occupations $(m_1, m_2 | n_1, n_2)$:



Their couplings are given by Kronecker structure

$$\left[\begin{array}{c} m_1 \ n_1 \\ \text{Y-shape} \\ m_2 \ n_2 \end{array} \right] = \delta_{m_2, n_2} \delta_{m_1-1, n_1+1} + \delta_{m_2, n_2} n_1 \delta_{m_1-1, n_1-1} \\ - \delta_{m_2, n_2+1} \delta_{m_1-1, n_1+1} - \delta_{m_2, n_2-1} n_2 \delta_{m_1-1, n_1}.$$

One-loop (warm-up).

$$\begin{array}{c} 2 \quad 1 \\ \text{---} \quad \text{---} \\ 1 \quad 1 \\ | \quad | \\ 0 \quad 0 \end{array} = \frac{1}{-\hat{i}\omega + r + k_o} \frac{1}{-\hat{i}\omega + r + k_o} \nu \begin{array}{c} 1 \quad 1 \\ \text{Y-shape} \\ 0 \quad 0 \end{array}$$

Two-loop.

$$\begin{array}{c} \text{---} \quad \text{---} \quad \text{---} \quad \text{---} \quad \text{---} \\ | \quad | \quad | \quad | \quad | \\ \text{---} \quad \text{---} \quad \text{---} \quad \text{---} \quad \text{---} \end{array} = \frac{1}{-\hat{i}\omega + r + k_o} \frac{1}{-\hat{i}\omega + r + k_o} \sum_{\ell_1, \ell_2} \nu \begin{array}{c} 1 \quad \ell_1 \\ \text{Y-shape} \\ 0 \quad \ell_2 \end{array} \nu \left(\begin{array}{c} \ell_1 \quad 1 \\ \text{Y-shape} \\ \ell_2 \quad 0 \end{array} + \begin{array}{c} \ell_2 \quad 0 \\ \text{Y-shape} \\ \ell_1 \quad 1 \end{array} \right) \\ \times \underbrace{\int d\omega' \frac{1}{-\hat{i}(\omega - \omega') + r + \ell_1 k_o} \frac{1}{-\hat{i}\omega' + r + \ell_2 k_o}}_{\frac{1}{-\hat{i}\omega + 2r + (\ell_1 + \ell_2)k_o}}$$

By Mathematica:

$$= \frac{1}{-\hat{i}\omega + r + k_o} \frac{1}{-\hat{i}\omega + r + k_o} \nu^2 \frac{2}{-\hat{i}\omega + 2r + k_o}.$$

Three-loop.

$$\begin{aligned}
 \text{Diagram} &= \frac{1}{-\hat{i}\omega + r + k_o} \frac{1}{-\hat{i}\omega + r + k_o} \sum_{\ell_1 \dots \ell_4} \nu \begin{array}{c} \ell_1 \\ \diagup \quad \diagdown \\ 0 \quad \ell_2 \end{array} \nu \\
 &\times \left(\begin{array}{c} \ell_1 \quad \ell_3 \\ \diagup \quad \diagdown \\ \ell_2 \quad \ell_4 \end{array} + \begin{array}{c} \ell_2 \quad \ell_4 \\ \diagup \quad \diagdown \\ \ell_1 \quad \ell_3 \end{array} \right) \left(\begin{array}{c} \ell_3 \quad 1 \\ \diagup \quad \diagdown \\ \ell_4 \quad 0 \end{array} + \begin{array}{c} \ell_4 \quad 0 \\ \diagup \quad \diagdown \\ \ell_3 \quad 1 \end{array} \right) \\
 &\underbrace{\hspace{10em}}_{\frac{1}{-\hat{i}\omega + 2r + (\ell_1 + \ell_2)k_o}} \underbrace{\hspace{10em}}_{\frac{1}{-\hat{i}\omega + 2r + (\ell_3 + \ell_4)k_o}}
 \end{aligned}$$

By Mathematica:

$$= \frac{1}{-\hat{i}\omega + r + k_o} \frac{1}{-\hat{i}\omega + r + k_o} \nu^3 \frac{4}{(-\hat{i}\omega + 2r + k_o)^2}. \quad \text{Hm.}$$

General n -loop sum. So we need

$$\begin{aligned}
 \sum_{\ell_1, \dots, \ell_{2n}} \begin{array}{c} 1 \quad \ell_1 \\ \diagup \quad \diagdown \\ 0 \quad \ell_2 \end{array} \cdot \left(\begin{array}{c} \ell_1 \quad \ell_3 \\ \diagup \quad \diagdown \\ \ell_2 \quad \ell_4 \end{array} + \begin{array}{c} \ell_2 \quad \ell_4 \\ \diagup \quad \diagdown \\ \ell_1 \quad \ell_3 \end{array} \right) \left(\begin{array}{c} \ell_3 \quad \ell_5 \\ \diagup \quad \diagdown \\ \ell_4 \quad \ell_6 \end{array} + \begin{array}{c} \ell_4 \quad \ell_6 \\ \diagup \quad \diagdown \\ \ell_3 \quad \ell_5 \end{array} \right) \cdots \\
 \left(\begin{array}{c} \ell_{2n} \ell_{2n-1} \\ \diagup \quad \diagdown \\ \ell_{2n} \ell_{2n} \end{array} + \begin{array}{c} \ell_{2n} \ell_{2n} \\ \diagup \quad \diagdown \\ \ell_{2n} \ell_{2n-1} \end{array} \right) \left(\begin{array}{c} \ell_{2n-1} \\ \diagup \quad \diagdown \\ \ell_{2n} \quad 0 \end{array} + \begin{array}{c} \ell_{2n} \quad 0 \\ \diagup \quad \diagdown \\ \ell_{2n-1} \end{array} \right) = 2^n.
 \end{aligned}$$

(I can compute that using a very different route.)

How can I see that?
Generating functions? Really?

Response / Commentary

Reading Gunnar's note

The note is a sharpened restatement of his original challenge with two distinct charges:

Charge 1 (philosophical, §1). Standard φ^4 exponents do not come from *asymptotic* diagram counts. They come from a finite ratio of two diagrams, with their finite Wick combinatorics ($4 \cdot 3$ for the tadpole; $\binom{4}{2}^2 \cdot 2 \cdot 4! / 2$ for the bubble). The asymptotic-combinatorics framing of our Gribov document is therefore not the right lens for *this* problem — it's a counting problem at $O(1)$, not $O(n!)$.

Charge 2 (technical, §2). He hands us an explicit DP problem: an H -vertex algebra with the four-term Kronecker coupling, and a chain diagram whose n -loop sum — evaluated in Mathematica through three loops — gives exactly 2^n . He is asking: *can a generating-function / asymptotic-combinatorics framework see this 2^n structurally, or is it just a brute Mathematica result?*

R1. On Charge 1 — not a concession: AC *is* the counting machine

Gunnar is right that the φ^4 exponents come from *counting*, not from $n!^\alpha$ asymptotics. But “counting” and “asymptotic combinatorics” are not different machines — they are the same generating function evaluated at different orders. The word “asymptotic” was a poor framing in the Gribov draft; the underlying object is a generating function $C(z) = \sum_n c_n z^n$ over labelled-diagram weights, and:

- the low-order *coefficients* c_n are exactly the Wick weights $4 \cdot 3$ (tadpole) and $\binom{4}{2}^2 \cdot 2 \cdot 4!/2$ (bubble) that he highlights;
- the asymptotic growth of c_n (radius of convergence, $n!^\alpha$ prefactor) is what controls Tier-B observables like LERW $5/4$, TAP 1.624 , KPZ replica.

So AC supplies the same numerator/denominator combinatorics he is doing by hand — it is just *closed-form* rather than diagram-by-diagram. Concretely, the AC route to his §1 calculation is:

1. Write the labelled-diagram generating function $\mathcal{Z}(g, r)$ for φ^4 at one loop. Its g^1 coefficient is automatically the symmetry-weighted sum of the two diagrams.
2. Read off the Wick weights as derivatives: $c_{\text{tadpole}} = [g^1] \partial_r \mathcal{Z}$ gives $4 \cdot 3$; $c_{\text{bubble}} = [g^2] \partial_g \mathcal{Z}$ gives $\binom{4}{2}^2 \cdot 2 \cdot 4!/2$. No hand-counting of Wick contractions.
3. The ratio of these coefficients is a generating-function residue — it is what enters β_u , and it produces $g^* = \varepsilon/36$, $\nu = \frac{1}{2} + \frac{\varepsilon}{12}$ on the nose.

In other words: AC contains standard φ^4 exponent-extraction as the leading-order truncation of its own generating function. The Tier-B observables are the same machine run to all orders, where the $n \rightarrow \infty$ asymptotics of the same coefficients carry the additional information.

The right reframing, then, is not “asymptotic combinatorics is a different problem from φ^4 ” but “ φ^4 exponents are the low-order residue of the same generating function whose $n \rightarrow \infty$ tail gives Tier-B.” We should rewrite the Gribov framing in those terms — and that, incidentally, is also the framework that answers Charge 2 below.

R2. On Charge 2 — the 2^n is structural, and a generating function does see it

The 2^n is exactly what a generating-function argument should produce. Two converging observations:

(a) **The factor of 2 per vertex pair is the symmetry of the rung.** At each rung he has

two terms in the bracket, $\sum_{\ell_{2k} \ell_{2k+1}} + \sum_{\ell_{2k} \ell_{2k+2}}$, which is the symmetrisation over the two horizontal

lines of the ladder. The contraction algebra of his four-term coupling (two $\delta\delta$ pieces, two $n_i \delta\delta$ pieces) has the property that each rung’s bracket evaluates to twice a single oriented contribution after the $\sum_\ell$ collapses the δ -structure. With n rungs that is 2^n .

(b) **Generating function.** Encode each leg occupation ℓ by a marker z^ℓ . The four-term coupling becomes a 2×2 transfer matrix $T(z)$ acting on the pair of “rail states” (top, bottom); the bracket symmetrisation makes T act as $T + T^\top$ on each rung. The chain contraction is then

$$\langle \text{boundary}_L | (T + T^\top)^n | \text{boundary}_R \rangle.$$

The four-term Kronecker structure is engineered so that $T + T^\top$ has *eigenvalue 2* on the boundary subspace (the only sector the boundary projects onto), giving 2^n exactly. This is not asymptotics — it is a 1×1 eigenvalue computation on a transfer matrix the generating function delivers in closed form.

A The actual calculations

A.1 A. φ^4 Wick weights as generating-function coefficients

Take the labelled-source generating function

$$\mathcal{Z}(g, r; J) = \int \mathcal{D}\varphi \exp\left\{-\frac{1}{2} \int (\nabla\varphi)^2 - \frac{r}{2} \int \varphi^2 - g \int \varphi^4 + \int J\varphi\right\}.$$

By construction, expanding $\mathcal{Z}$ in g produces exactly the labelled-graph sum with Wick weights:

$$\mathcal{Z}(g, r; J) = \mathcal{Z}_0[J] \sum_{k \geq 0} \frac{(-g)^k}{k!} \left\langle \left(\int \varphi^4 \right)^k \right\rangle_{0, J}.$$

Tadpole (g^1 , two-point function). The one-loop self-energy is the g^1 -coefficient of $-\partial_{J(x)} \partial_{J(y)} \log \mathcal{Z}$ at $J = 0$:

$$[g^1] \Sigma(x, y) = -\langle \varphi(x) \varphi(y) \int dz \varphi(z)^4 \rangle_{0, \text{conn}} = -12 \int dz G(x-z) G(0) G(z-y).$$

The number **12** = $4 \cdot 3$ falls out of the Gaussian moment as Wick pairings of 6 fields: 4 ways for $\varphi(x)$ to land on a z -leg, 3 for $\varphi(y)$ on a remaining z -leg, the last two z -legs make the loop. *Identical to Gunnar's 4 · 3 in §1(a)* — but here it is read off as a Gaussian-moment derivative, no hand contractions.

Bubble (g^2 , four-point function). Same machine, one order up:

$$[g^2] \Gamma^{(4)} = \frac{1}{2!} \int dz_1 dz_2 \langle \varphi^4(z_1) \varphi^4(z_2) \rangle_{1\text{PI}, 4\text{-ext}} = \left(\frac{4}{2}\right)^2 \cdot 2 \cdot 4! \cdot \frac{1}{2} \int \frac{d^d k}{(k^2 + r)^2}.$$

The combinatorial factor decomposes as Gunnar wrote it: $\left(\frac{4}{2}\right)^2$ for the loop-leg choice at each vertex, 2 for matching the two internal lines, $4!$ for assigning external lines to slots, $\frac{1}{2}$ for the vertex-permutation symmetry. *Identical to Gunnar's §1(b).*

Resulting exponent. Feeding these into β_u and Z_r at the $g^* = \varepsilon/36$ fixed point gives $\nu = \frac{1}{2} + \frac{\varepsilon}{12}$ on the nose, by his own algebra. *The point is that AC produced his combinatorial factors mechanically, not that we are doing a different calculation.*

A.2 B. The 2^n ladder from a 2×2 transfer matrix

Restrict Gunnar's H-vertex coupling to its action on the "boundary subspace" $\mathcal{B} = \{|(1, 0)\rangle, |(0, 1)\rangle\}$. Term-by-term on $|(1, 0)\rangle$: only Term 2 ($m_1 = n_1 = 1, m_2 = n_2 = 0$, weight $n_1 = 1$) and Term 4 ($n_2 = 1, n_1 = 0$, weight $-n_2 = -1$) survive — the others demand negative occupations. So

$$V|(1, 0)\rangle = +|(1, 0)\rangle - |(0, 1)\rangle, \quad V|(0, 1)\rangle = 0.$$

Identical analysis for the rail-flipped vertex $V^\top(m_1, m_2; n_1, n_2) \equiv V(m_2, m_1; n_2, n_1)$:

$$V^\top|(1, 0)\rangle = 0, \quad V^\top|(0, 1)\rangle = +|(0, 1)\rangle - |(1, 0)\rangle.$$

On $\mathcal{B}$ in the basis $\{(1, 0), (0, 1)\}$:

$$V = \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix}, \quad V^\top = \begin{pmatrix} 0 & 0 \\ -1 & 1 \end{pmatrix}, \quad \boxed{T \equiv V + V^\top = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}.$$

The matrix T is rank-1 with eigenvalues $\{0, 2\}$ and satisfies

$$T^2 = \begin{pmatrix} 2 & -2 \\ -2 & 2 \end{pmatrix} = 2T \implies T^n = 2^{n-1}T \quad (n \geq 1).$$

Chain evaluation. Gunnar's n -bracket ladder ($n + 1$ H-vertices, leftmost fixed as V , the rest paired as $V + V^\top$) is the matrix element

$$\langle (1,0) | V T^n | (1,0) \rangle = 2^{n-1} \langle (1,0) | VT | (1,0) \rangle.$$

Direct computation:

$$VT = \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & -2 \\ 0 & 0 \end{pmatrix}, \quad (VT)_{(1,0),(1,0)} = 2.$$

Therefore

$$\langle (1,0) | V T^n | (1,0) \rangle = 2^{n-1} \cdot 2 = \boxed{2^n}.$$

No Mathematica. The factor of 2 per rung is the eigenvalue of the rung's symmetrised transfer matrix on the boundary sector — structural, not asymptotic.

Spot checks. $n = 1$ (his 2-loop ladder): $2^1 = 2$. $n = 2$ (his 3-loop ladder): $2^2 = 4$. $n = 0$ (single rung, no bracket): $\langle (1,0) | V | (1,0) \rangle = 1 = 2^0$.

Why the boundary sector is closed. Total occupation $m_1 + m_2$ is not conserved by V in general (Terms 1, 3 add particles), but on states with $m_1 + m_2 = 1$ those terms require negative outputs and so are killed. The dynamics therefore close on $\mathcal{B}$, which is why a 2×2 matrix suffices.

Generating-function gloss. Encoding V as a polynomial $\hat{V}(z_1, z_2; w_1, w_2) = \sum V(m; n) z_1^{m_1} z_2^{m_2} w_1^{n_1} w_2^{n_2}$ turns the four-term Kronecker structure into a closed-form rational function; the 2×2 transfer matrix is its restriction to the linear sector $\{z_1, z_2\} \otimes \{w_1, w_2\}$. The eigenvalue 2 is $\hat{V}(1, 1; 1, 1)$ projected onto the antisymmetric rail mode.

What to send back

A short note that:

1. *Reframes* Charge 1 (not concedes it): AC is the generating-function machine whose low-order coefficients are exactly his $4 \cdot 3$ and $\binom{4}{2}^2 \cdot 2 \cdot 4! / 2$. The "asymptotic" label was wrong; the machine is the same one φ^4 already uses, written closed-form. Show this by reproducing his $\nu = \frac{1}{2} + \frac{\epsilon}{12}$ from a one-line generating-function residue.
2. On Charge 2, sets up the transfer matrix $T(z)$ from the four-term coupling, computes $T + T^\top$ on the boundary sector, and shows the eigenvalue is exactly 2 — so the chain sum is 2^n on the nose, no Mathematica needed.
3. Then exhibits one DP observable (LERW 5/4 or TAP 1.624) where the same generating function, run to its $n \rightarrow \infty$ tail, is the *only* known route — to show the two regimes are endpoints of one machine, not two different machines.